

Rowzatul Zannath Prerona

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Bachelor of Science

Ahsanullah University of Science and Technology, Dhaka

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🐙 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

- **Bachelor of Science in Computer Science and Engineering** 2022-26
Ahsanullah University of Science and Technology, Dhaka CGPA: 3.46
- **Higher Secondary Certificate** 2018-20
Holy Cross College, Dhaka GPA: 5.00
- **Secondary School Certificate** 2015-18
Ideal School and College, Motijheel, Dhaka GPA: 5.00

PERSONAL PROJECTS

•EfficientNet-CBAM: Attention-Guided Brain Tumor Classification

An attention-guided deep learning framework for multi-class brain tumor classification from MRI scans.

- Integrated Convolutional Block Attention Module (CBAM) into an EfficientNetB3 backbone to focus on pathologically relevant regions, achieving **99.29%** classification accuracy.
- Implemented HiResCAM gradient-based visualizations for model explainability, improving clinical interpretability across Glioma, Meningioma, and Pituitary Tumor classes.
- Technology Used: Python, TensorFlow, Keras, EfficientNetB3, CBAM, HiResCAM, OpenCV, NumPy, scikit-learn.

•Deforestation Mapping in Dhaka — Semantic Segmentation

Deep learning-based semantic segmentation to detect and monitor deforested areas using satellite imagery.

- Built a high-resolution satellite imagery dataset of Dhaka spanning 2012–2022 and benchmarked three architectures: U-Net, Attention U-Net, and a hybrid ResNet50-Attention U-Net.
- Evaluated models using Dice Score, IoU, F1-Score, Precision, and Recall; providing a reproducible framework for environmental change monitoring.
- Technology Used: Python, TensorFlow, Keras, PyTorch, U-Net, ResNet50, NumPy, OpenCV, scikit-learn, matplotlib.

•Planet Paw — Animal Rescue & Welfare Platform

A multifaceted web platform integrating donation processing, pet adoption, medical aid listings, and food sales.

- Designed a community-driven ecosystem supporting full CRUD operations for rescue listings, donor management, and inventory with emphasis on transparency and sustainability.
- Built with a PHP/MySQL backend managed via XAMPP and phpMyAdmin, delivering seamless user engagement across multiple welfare services.
- Technology Used: PHP, HTML, CSS, JavaScript, MySQL, XAMPP, phpMyAdmin.

•LeafLedger — Bookshop Management System

A full-stack bookshop management system streamlining inventory, sales, and customer operations.

- Implemented book and genre management, shopping cart functionality, stock monitoring, order processing, and admin-level operations including book approvals and content management.
- Built on ASP.NET Core MVC with a Bootstrap frontend and SQL Server database, providing a clean administrative interface suitable for real-world retail deployment.
- Technology Used: ASP.NET Core MVC, Bootstrap, HTML, CSS, SQL Server, .NET 8 SDK, SSMS.

•Spread Love for Pet — Pet Management Desktop Application

A feature-rich Java desktop application consolidating pet management, health monitoring, and community engagement.

- Developed user and pet profile modules, an expense tracker, a pet health encyclopedia, a BMI calculator, and a community connectivity module for pet owners.
- Technology Used: Java.

•Paw Prints Manager — Pet Care Mobile Application

A comprehensive all-in-one pet care mobile application built with Flutter for pet owners.

- Features include pet and user profile management, a monthly expense tracker, a health encyclopedia searchable by concern, a BMI calculator, and a reminder system for vet visits and medications.
- Technology Used: Flutter, Dart.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, C/C++, JavaScript, HTML + CSS, Dart, Java

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Flutter, ReactJs, ASP.NET Core MVC, Bootstrap

ML/DL & CV Tools: OpenCV, NumPy, scikit-learn, matplotlib, CBAM, HiResCAM, EfficientNetB3, ResNet50, U-Net

Databases: MySQL, SQL Server, Firebase, MongoDB

Dev Tools: Git, GitHub, VS Code, XAMPP, phpMyAdmin, SSMS, .NET 8 SDK

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Database Management System, Operating Systems, Software Engineering

Areas of Interest: Medical Image Processing, Deep Learning, Computer Vision, AI in Healthcare, Semantic Segmentation.

Soft Skills: Research-driven Problem Solving, Self-learning, Adaptability, Presentation